

PCP BC - Capstone Project

Course-end Project 2: Intellectual Property Protection with Blockchain

Name: Neeleshwari Mehra

Project Description:

Objectives

To develop blockchain-based systems for registering, managing, and verifying intellectual property rights ownership using decentralised technology.

Problem Statement and Motivation

Problem Statement:

Intellectual property (IP) rights are critical for creators and innovators to protect their inventions, creations, and designs. However, the current systems for registering and managing IP rights often face challenges related to proof of ownership, preventing unauthorized use, and ensuring transparency in transactions. Blockchain technology offers a decentralized and secure platform to address these issues effectively.

Real-World Scenario:

The patent office of a developing country depends on an outdated method for IP registration and management. The current system struggles with proving ownership, preventing unauthorized use, and ensuring transparent and tamper-proof records. They implement a blockchain solution that creates immutable records, decentralized verification, and enhanced security for intellectual property rights.

Project: Intellectual Property Protection with Blockchain

1. REMIX IDE CODE:

```
// SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;

contract IPProtection {

    enum IPType {

        Patent,

        Copyright,

        Trademark

    }

    struct IntellectualProperty {

        string title;

        string description;

        string ipHash;

        string ipType;

        address owner;

        uint256 timestamp;

    }

    mapping(uint256 => IntellectualProperty) public ipRecords;

    uint256 public ipCount;

    event IPRegistered(

        uint256 ipId,
```

```

        string title,
        string ipType,
        address owner
    );

    function registerIP(
        string memory _title,
        string memory _description,
        string memory _ipHash,
        IPType _type
    ) public {
        string memory typeString = getTypeString(_type);

        ipCount++;

        ipRecords[ipCount] = IntellectualProperty(
            _title,
            _description,
            _ipHash,
            typeString,
            msg.sender,
            block.timestamp
        );

        emit IPRegistered(ipCount, _title, typeString, msg.sender);
    }

    function showIPTypes()

```

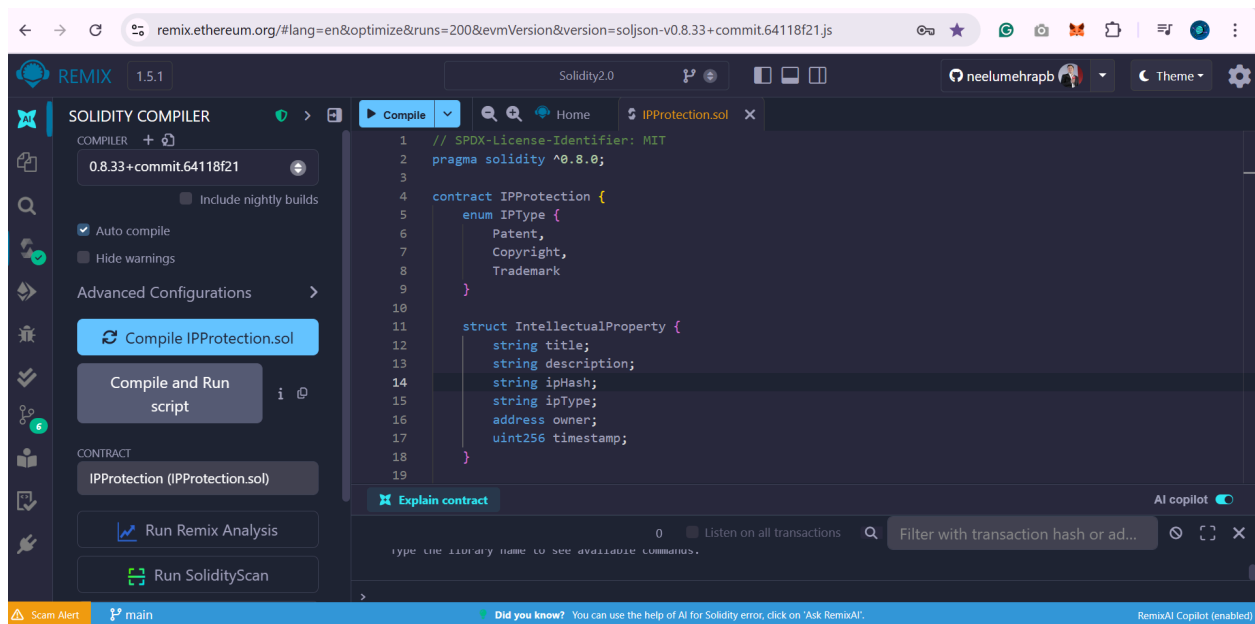
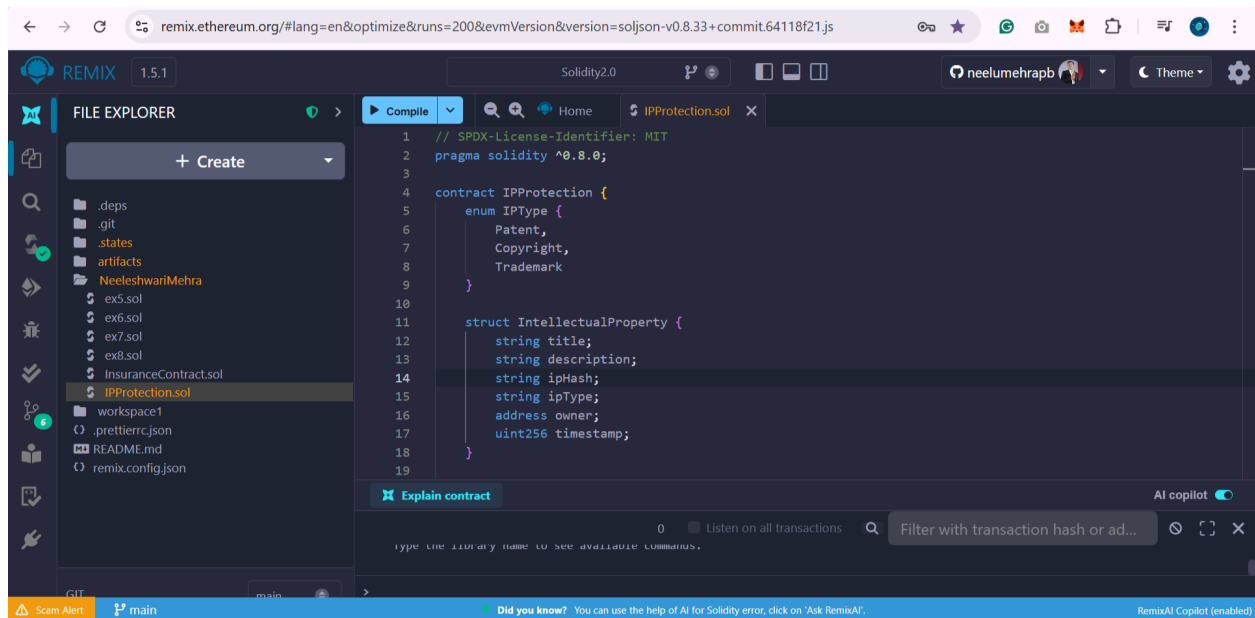
```
public
    pure
    returns (string memory, string memory, string memory)
{
    return ("0 - Patent", "1 - Copyright", "2 - Trademark");
}

function getTypeString(IPType _type) internal pure returns (string
memory) {
    if (_type == IPType.Patent) return "Patent";
    if (_type == IPType.Copyright) return "Copyright";
    return "Trademark";
}

function verifyIP(
    uint256 _ipId
)
public
view
returns (
    string memory,
    string memory,
    string memory,
    string memory,
    address,
    uint256
)
```

```
    {  
        IntellectualProperty memory ip = ipRecords[_ipId];  
  
        return (  
            ip.title,  
            ip.description,  
            ip.ipHash,  
            ip.ipType,  
            ip.owner,  
            ip.timestamp  
        );  
    }  
}
```

2. SCREENSHOT FOR DEPLOYMENT AND DEBUG OUTPUT:



REMIX1.5.1

Solidity2.0

neelumehrapb

Theme

DEPLOY & RUN TRANSACTIONS

ENVIRONMENT

Remix VM (Osaka)

ACCOUNT

0x5B3...eddC4 (100.0 ETH)

+ Authorize Delegation

GAS LIMIT

Estimated Gas

Custom3000000

VALUE

0Wei

CONTRACT

IPProtection - NeeleshwariMehra/IPP

evm version: osaka

Compile

Home

IPProtection.sol

1// SPDX-License-Identifier: MIT

2pragma solidity ^0.8.0;

3

4contract IPProtection {

5enum IPType {

6Patent,

7Copyright,

8Trademark

9}

10

11struct IntellectualProperty {

12string title;

13string description;

14string ipHash;

15string ipType;

16address owner;

17uint256 timestamp;

18}

19

Explain contract

AI copilot

0Listen on all transactions

Filter with transaction hash or ad...

Scam Alert

main

Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.

RemixAI Copilot (enabled)

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion=soljson-v0.8.33+commit.64118f21.js

REMIX1.5.1

Solidity2.0

neelumehrapb

Theme

DEPLOY & RUN TRANSACTIONS

ENVIRONMENT

Remix VM (Osaka)

ACCOUNT

0xe10...f1323 (0 ETH)

+ Authorize Delegation

GAS LIMIT

Estimated Gas

Custom3000000

VALUE

0Wei

CONTRACT

IPProtection - NeeleshwariMehra/IPP

evm version: osaka

Compile

Home

IPProtection.sol

1// SPDX-License-Identifier: MIT

2pragma solidity ^0.8.0;

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13string description;

14string ipHash;

15string ipType;

16address owner;

17uint256 timestamp;

18}

19

Explain contract

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0Listen on all transactions

Filter with transaction hash or ad...

Scam Alert

main

Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.

RemixAI Copilot (enabled)

0xe1...1323

0.199 ETH

Buy crypto

Swap

Send

Activity

Disconnect

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.33+commit.64118f21.js

REMIX 1.5.1 Solidity2.0 neelumehrapb Theme

DEPLOY & RUN TRANSACTIONS

ENVIRONMENT
WalletConnect
Sepolia (11155111) network
ACCOUNT
0xe10...f1323 (0.199519429817)
+ Create Smart Account

GAS LIMIT
Estimated Gas
Custom 3000000
VALUE
0 Wei
CONTRACT
IPProtection - NeeleshwariMehra/IPProtection
evm version: osaka

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract IPProtection {
5     enum IPType {
6         Patent,
7         Copyright,
8         Trademark
9     }
10
11     struct IntellectualProperty {
12         string title;
13         string description;
14         string ipHash;
15         string ipType;
16         address owner;
17         uint256 timestamp;
18     }
19 }
```

Explain contract

0 Listen on all transactions Filter with transaction hash or address

Scam Alert main Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'. RemixAI Copilot (enabled)

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.33+commit.64118f21.js

REMIX 1.5.1 Solidity2.0 neelumehrapb Theme

DEPLOY & RUN TRANSACTIONS

Custom 3000000
VALUE
0 Wei
CONTRACT
IPProtection - NeeleshwariMehra/IPProtection
evm version: osaka
Verify Contract on Explorers
Deploy
At Address Load contract from Address

Transactions recorded 1 i
Deployed Contracts 0

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract IPProtection {
5     enum IPType {
6         Patent,
7         Copyright,
8         Trademark
9     }
10
11     struct IntellectualProperty {
12         string title;
13         string description;
14         string ipHash;
15         string ipType;
16         address owner;
17         uint256 timestamp;
18     }
19 }
```

Explain contract

0 Listen on all transactions Filter with transaction hash or address

Scam Alert main Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'. RemixAI Copilot (enabled)

Account 1

Deploy a contract

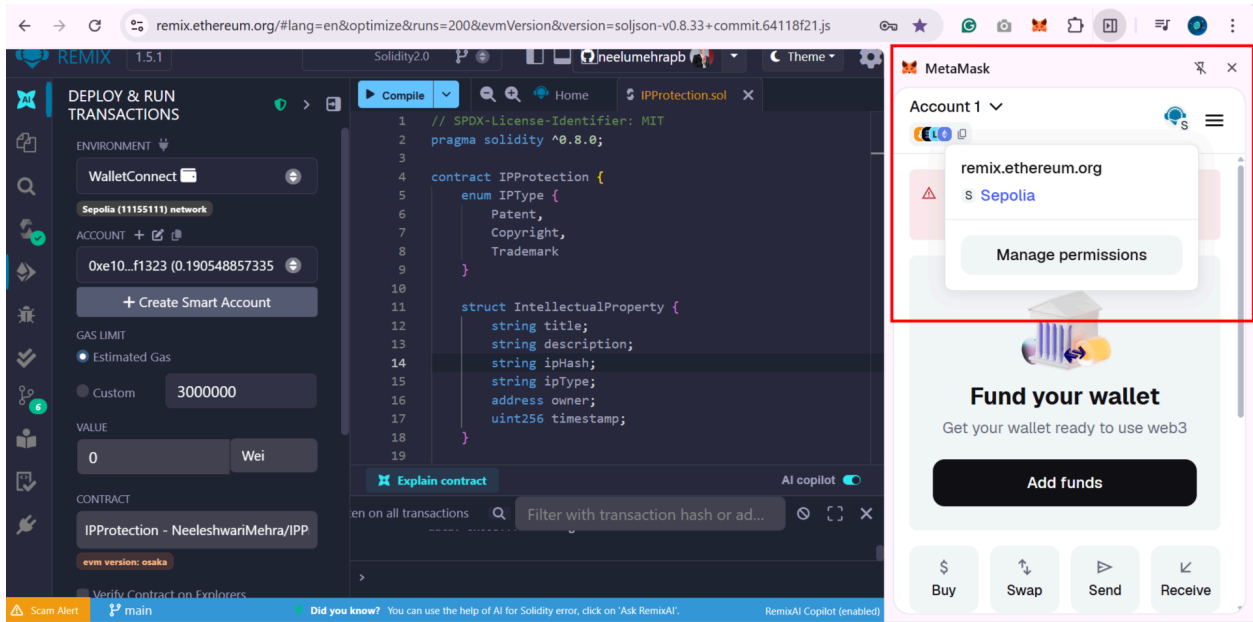
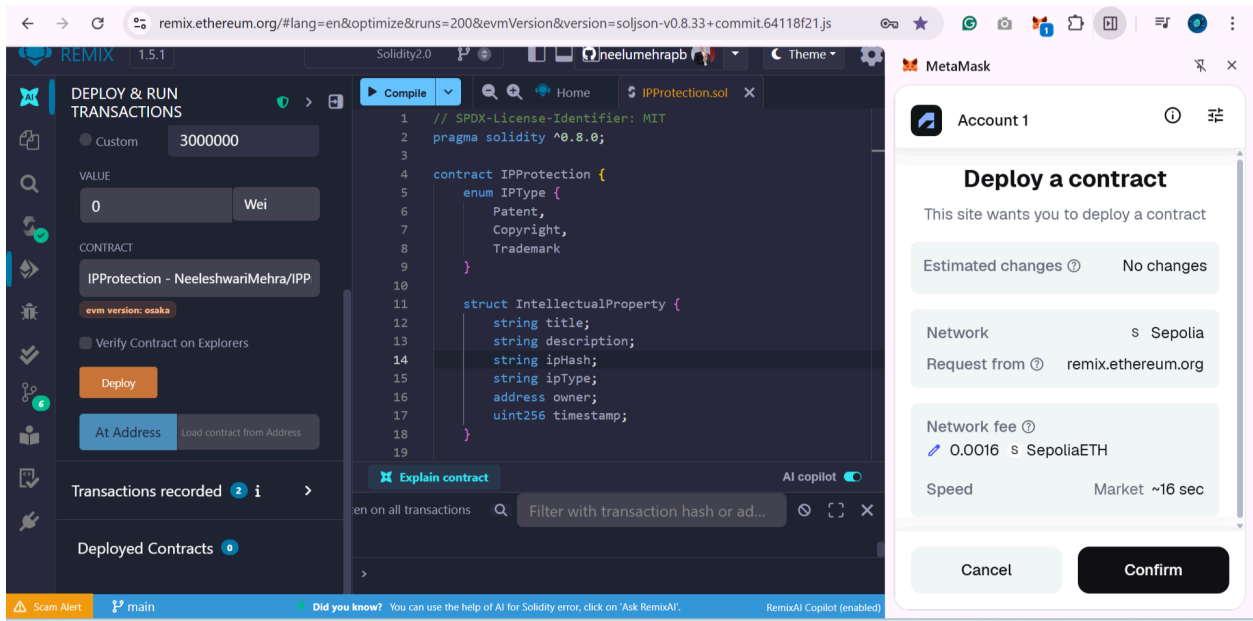
This site wants you to deploy a contract

Estimated changes No changes

Network s Sepolia
Request from remix.ethereum.org

Network fee 0.0017 s SepoliaETH
Speed Market ~16 sec

Cancel Confirm



remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.33+commit.64118f21.js

REMIX 1.5.1 Solidity2.0 neelumehrapb Theme

DEPLOY & RUN TRANSACTIONS

Deployed Contracts 1

IPPROTECTION AT 0XB1A...C00

Balance: 0 ETH

registerIP string_title, string_descript

ipCount

ipRecords uint256

showIPTypes

verifyIP uint256, _ipId

Low level interactions

CALLDATA

Transact

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract IPProtection {
5     enum IPType {
6         Patent,
7         Copyright,
8         Trademark
9     }
10
11     struct IntellectualProperty {
12         string title;
13         string description;
14         string ipHash;
15         string ipType;
16         address owner;
17         uint256 timestamp;
18     }
19 }
```

Explain contract AI copilot

Filter with transaction hash or ad...

Scam Alert main Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'. RemixAI Copilot (enabled)

MetaMask Account 1

remix.ethereum.org

Sepolia

Manage permissions

Fund your wallet

Get your wallet ready to use web3

Add funds

Buy Swap Send Receive

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.33+commit.64118f21.js

REMIX 1.5.1 Solidity2.0 neelumehrapb Theme

DEPLOY & RUN TRANSACTIONS

Calldata Parameters Transact

ipCount

ipRecords uint256

showIPTypes

0: string: 0 - Patent

1: string: 1 - Copyright

2: string: 2 - Trademark

verifyIP uint256, _ipId

Low level interactions

CALLDATA

Transact

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract IPProtection {
5     enum IPType {
6         Patent,
7         Copyright,
8         Trademark
9     }
10
11     struct IntellectualProperty {
12         string title;
13         string description;
14         string ipHash;
15         string ipType;
16         address owner;
17         uint256 timestamp;
18     }
19 }
```

Explain contract AI copilot

Filter with transaction hash or ad...

Scam Alert main Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'. RemixAI Copilot (enabled)

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remix.ethereum.org

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remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion=solcjs-v0.8.33+commit.64118f21.js

REMIX 1.5.1 Solidity 2.0

DEPLOY & RUN TRANSACTIONS
Deployed Contracts 1

IPPROTECTION AT 0xB1A...C007

Balance: 0 ETH

REGISTERIP

_title: mere bhole shankar

_description: shiv bhakti song

_ipHash: shiv12345

_type: 1

Calldata Parameters **transact**

ipCount

ipRecords: uint256

showIPTypes

string: 0 - Patent

string: 1 - Copyright

Compile

IPProtection.sol

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract IPProtection {
5     enum IPTYPE {
6         Patent,
```

Explain contract

AI copilot

Logs

raw logs

transact to IPProtection.registerIP pending ...

transact to IPProtection.registerIP errored: Error occurred: user rejected action

user rejected action (action="sendTransaction", reason="rejected", info={ "error"

If the transaction failed for not having enough gas, try increasing the gas limit

transact to IPProtection.registerIP pending ...

MetaMask

Account 1

Transaction request

Estimated changes No changes

Network s Sepolia

Request from remix.ethereum.org

Interacting with OxB1A6f...C007D

Network fee 0.0015 s SepoliaETH

Speed Market ~16 sec

Cancel Confirm

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion=solcjs-v0.8.33+commit.64118f21.js

REMIX 1.5.1 Solidity 2.0

DEPLOY & RUN TRANSACTIONS
Deployed Contracts 1

IPPROTECTION AT 0xB1A...C007

Balance: 0 ETH

REGISTERIP

_title: mere bhole shankar

_description: shiv bhakti song

_ipHash: shiv12345

_type: 1

Calldata Parameters **transact**

ipCount

ipRecords: uint256

showIPTypes

string: 0 - Patent

string: 1 - Copyright

Compile

IPProtection.sol

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.0;
3
4 contract IPProtection {
5     enum IPTYPE {
6         Patent,
```

Explain contract

AI copilot

Logs

raw logs

user rejected action (action="sendTransaction", reason="rejected", info={ "error"

If the transaction failed for not having enough gas, try increasing the gas limit

transact to IPProtection.registerIP pending ...

view on Etherscan view on Blockscout

transact to IPProtection.registerIP pending ...

[block:10246180 txIndex:28] from: 0xE10...f1323

to: IPProtection.registerIP(string,string,string,uint8)

0xB1A...C007D

value: 0 wei data: 0x7ec...00000 logs: 1

hash: 0x325...20a2b

Debug

MetaMask

Account 1

Transaction request

Estimated changes No changes

Network s Sepolia

Request from remix.ethereum.org

Interacting with OxB1A6f...C007D

Network fee 0.0012 s SepoliaETH

Speed Market ~16 sec

Cancel Confirm

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion=soljson-v0.8.33+commit.64118f21.js

REMIX 1.5.1

Solidity2.0

neelumehrapb

Theme

DEPLOY & RUN TRANSACTIONS

Deployed Contracts

IPPROTECTION AT 0xB1A...C00

Balance: 0 ETH

REGISTERIP

_title: mere bhole shankar

_description: shiv bhakti song

_ipHash: shiv12345

_type: 1

CallData

Parameters

transact

ipCount

ipRecords: uint256

showIPTypes

string: 0 - Patent

string: 1 - Copyright

Compile

Home

IPProtection.sol

1 // SPDX-License-Identifier: MIT

2 pragma solidity ^0.8.0;

3

4 contract IPProtection {

5 enum IPTYPE {

6 Patent,

7 }

8 }

Explain contract

AI copilot

Filter with transaction hash or address

block number: 10246183

from: 0xE18FF59a60C18E1D8372F31D36FEc442178f1323

to: IPProtection.registerIP(string,string,string,uint8)

transaction cost: 170315 gas

decoded input: { "string_title": "mere bhole shankar", "string_description": "shiv bhakti song", "string_ipHash": "shiv12345", "uint8_type": "1" }

MetaMask

Account 1

Unable to connect to simplilearn test1. Check network connectivity, or update RPC.

Fund your wallet

Get your wallet ready to use web3

Add funds

Buy Swap Send Receive

remix.ethereum.org/#lang=en&optimize&runs=200&evmVersion=soljson-v0.8.33+commit.64118f21.js

REMIX 1.5.1

Solidity2.0

neelumehrapb

Theme

DEPLOY & RUN TRANSACTIONS

Deployed Contracts

IPPROTECTION AT 0xB1A...C00

Balance: 0 ETH

REGISTERIP

_title: mere bhole shankar

_description: shiv bhakti song

_ipHash: shiv12345

_type: 1

CallData

Parameters

transact

ipCount

ipRecords: uint256

showIPTypes

string: 0 - Patent

string: 1 - Copyright

Compile

Home

IPProtection.sol

1 // SPDX-License-Identifier: MIT

2 pragma solidity ^0.8.0;

3

4 contract IPProtection {

5 enum IPTYPE {

6 Patent,

7 }

8 }

Explain contract

AI copilot

Filter with transaction hash or address

view on Etherscan view on Blockscout

[block:10246183 txIndex:22] from: 0xE10...f1323

to: IPProtection.registerIP(string,string,string,uint8)

0xB1A...C007D

value: 0 wei data: 0x7ec...0000 logs: 1

hash: 0x068...31f79

status: 1 Transaction mined and execution succeed

transaction hash: 0xbfe2abc052f0c1fe5412c90f4907b8536af8080d03f9c7982923

671d00ba49ee

block hash: 0x068c8af3c66b1bfeb5be58ea3b569c904e785d7beeac0bf570f

ebfb94731f79

block number: 10246183

Debug

MetaMask

Account 1

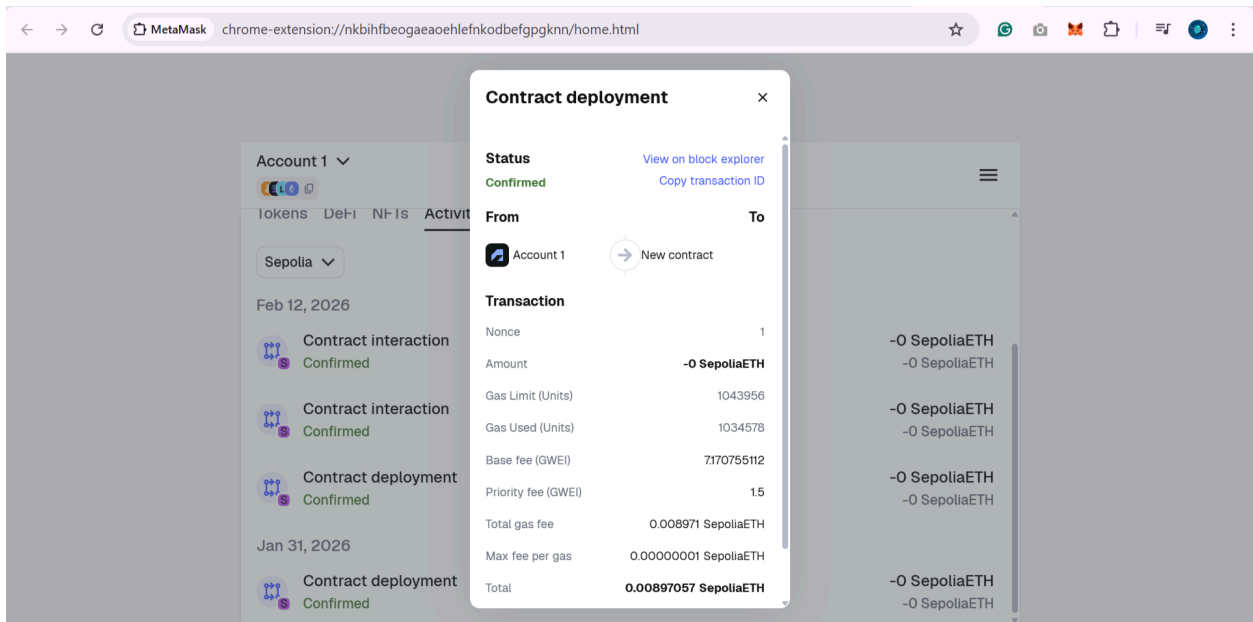
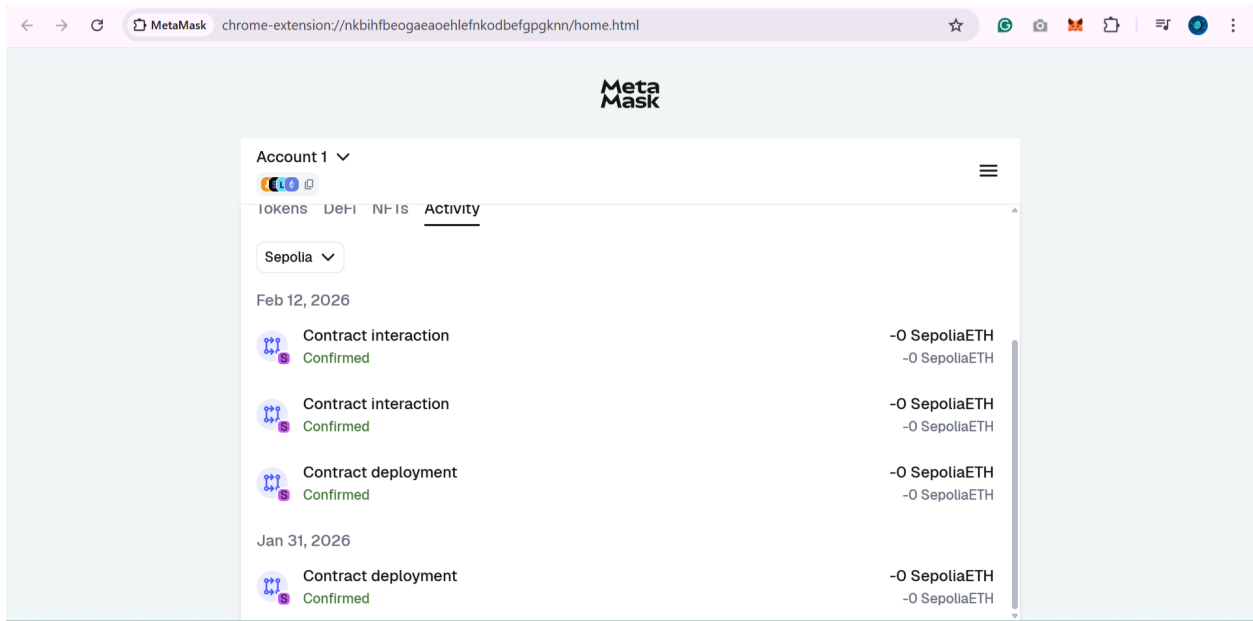
Unable to connect to simplilearn test1. Check network connectivity, or update RPC.

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Get your wallet ready to use web3

Add funds

Buy Swap Send Receive



MetaMask chrome-extension://nkbihfbeogaeaoehlefnkodbefgpgknn/home.html

Account 1

Tokens DeFi NFTs Activity

Sepolia

Feb 12, 2026

Contract interaction

Confirmed

Contract interaction

Confirmed

Contract deployment

Confirmed

Jan 31, 2026

Contract deployment

Confirmed

Contract interaction

Confirmed

[View on block explorer](#)
[Copy transaction ID](#)

From

Account 1

To

OxB1A6f...C007D

Transaction

Nonce

2

Amount

-0 SepoliaETH

Gas Limit (Units)

190070

Gas Used (Units)

187415

Base fee (GWEI)

6190272003

Priority fee (GWEI)

1.5

Total gas fee

0.001441 SepoliaETH

Max fee per gas

0.00000001 SepoliaETH

Total

0.00144127 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

MetaMask chrome-extension://nkbihfbeogaeaoehlefnkodbefgpgknn/home.html

Account 1

Tokens DeFi NFTs Activity

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Feb 12, 2026

Contract interaction

Confirmed

Contract interaction

Confirmed

Contract deployment

Confirmed

Jan 31, 2026

Contract deployment

Confirmed

Contract interaction

Confirmed

[View on block explorer](#)
[Copy transaction ID](#)

From

Account 1

To

OxB1A6f...C007D

Transaction

Nonce

3

Amount

-0 SepoliaETH

Gas Limit (Units)

172834

Gas Used (Units)

170315

Base fee (GWEI)

5725096242

Priority fee (GWEI)

1.5

Total gas fee

0.001231 SepoliaETH

Max fee per gas

0.00000001 SepoliaETH

Total

0.00123054 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

-0 SepoliaETH

← → ×

sepolia.etherscan.io/tx/0xbfe2abc052f0c1fe5412c90f4907b8536af080d0d3f9c7982923671d00ba49ee

🔍 ☆ 🗺 📷 📄 📱 📡

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Etherscan

Home Blockchain Tokens NFTs More

Transaction Details

Overview Logs (1) State

</> API

TRANSACTION ACTION

Call 0x7eceb7a7 Method by 0xE10Ff59a...2178f1323 on 0xB1A6fE9d...950fC007D

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0xbfe2abc052f0c1fe5412c90f4907b8536af080d0d3f9c7982923671d00ba49ee

Status:

Success

Block:

10246183 50 Block Confirmations

Timestamp:

10 mins ago (Feb-12-2026 03:42:24 PM UTC)

From:

0xE10Ff59a60C18E1D8372F31D36FEc442178f1323

To:

0xB1A6fE9d16E321cAaBfA61f66B71eb9950fC007D

Value:

0 ETH

Transaction Fee:

0.00123054226645623 ETH

Gas Price:

7.225096242 Gwei (0.000000007225096242 ETH)

← → ×

sepolia.etherscan.io/tx/0x1932ad27c3ceccb9f0c622c6119c8d96ee0b7743e7cdb878c4c5aff1e632ac13

🔍 ☆ 🗺 📷 📄 📱 📡

Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Etherscan

Home Blockchain Tokens NFTs More

Transaction Details

Overview Logs (1) State

</> API

TRANSACTION ACTION

Call 0x7eceb7a7 Method by 0xE10Ff59a...2178f1323 on 0xB1A6fE9d...950fC007D

[This is a Sepolia Testnet transaction only]

Transaction Hash:

0x1932ad27c3ceccb9f0c622c6119c8d96ee0b7743e7cdb878c4c5aff1e632ac13

Status:

Success

Block:

10246180 61 Block Confirmations

Timestamp:

13 mins ago (Feb-12-2026 03:41:24 PM UTC)

From:

0xE10Ff59a60C18E1D8372F31D36FEc442178f1323

To:

0xB1A6fE9d16E321cAaBfA61f66B71eb9950fC007D

Value:

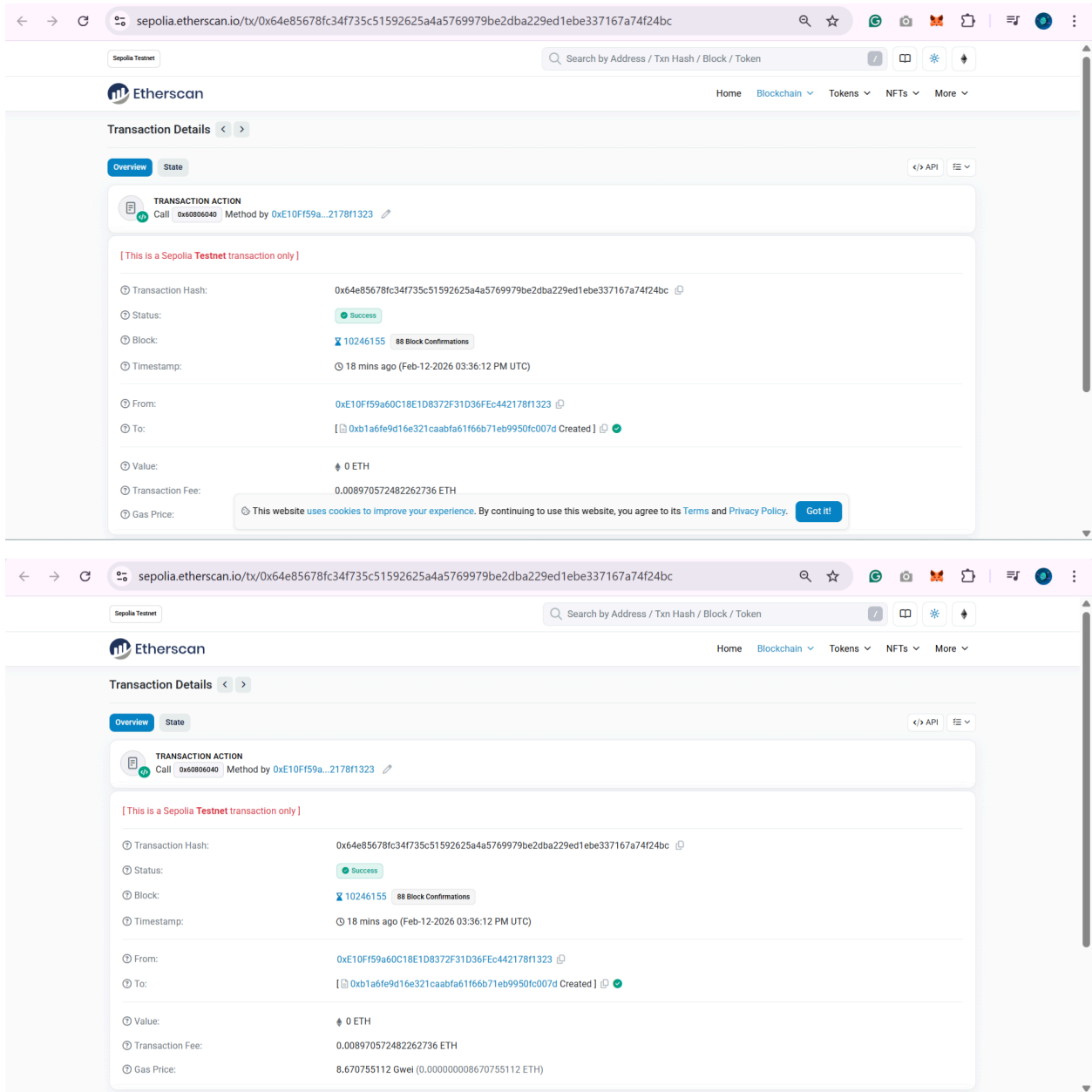
0 ETH

Transaction Fee:

0.0014417737447745 ETH

Gas Price:

This website uses cookies to improve your experience. By continuing to use this website, you agree to its Terms and Privacy Policy. Got it!



3. DEBUG OUTPUT:

Welcome to Remix 1.5.1

Your files are stored in indexedDB, 3.6 MB / 142.46 GB used

You can use this terminal to:

- Check transactions details and start debugging.

- Execute JavaScript scripts:
 - Input a script directly in the command line interface
 - Select a Javascript file in the file explorer and then run ``remix.execute()`` or ``remix.exeCurrent()`` in the command line interface
 - Right-click on a JavaScript file in the file explorer and then click ``Run``

- ethers.js

```
creation of IPProtection pending...
```

```
(action="sendTransaction", reason="rejected", info={ "error": { "code": 4001, "data": { "cause": "rejectAllApprovals" }, "message": "ethers-user-denied: MetaMask Tx Signature: User denied transaction signature." }, "payload": { "id": 35, "jsonrpc": "2.0", "method":
```

[illegible]


```
0043610610054575f3560e1c80621e5c1c1461005857806302c59bc61461008d57806311d3de74146100c25780637  
eeeb7a7146100e2578063907e67fe146100fe575b5f5ffd5610072600480360381019061006d5906109f5556b610  
11c565b60405161008496959493929190610ade565b60405180910390f35b6100a760048036038101906100a291906  
109f5556b610413565b6040516100b9965959493929190610ade565b60405180910390f35b6100ca610681565b60405  
16100d993929190610bf59565b60405180910390f35b6100fc60048036038101906100f79190610cf2565b610733565  
b005b61010661089a565b6040516101139190610daa565b60405180910390f35b60608060805f5f5f5f5f8981526  
020019081526020015f206040518060c00160405280915f8201805461014f90610df0565b80601f016020809104026  
020016040519081016040528092919081815260200182805461017b90610df0565b80156101c65780601f1061019d5  
76101008083540402835291602001916101c6565b820191905f5260205f20905b81548152906001019060200180831  
16101a957829003601f168201915b505050505081526020016001820180546101df90610df0565b80601f016020809  
104026020016040519081016040528092919081815260200182805461020b90610df0565b80156102565780601f10  
1022d57610100808354040283529160200191610256565b820191905f5260205f20905b81548152906001019060201  
180831161023957829003601f168201915b5050505050815260200160028201805461026f90610df0565b80601f016  
020809104026020016040519081016040528092919081815260200182805461029b90610df0565b80156102e657806  
01f106102bd576101008083540402835291602001916102e6565b820191905f5260205f20905b81548152906001019  
06020018083116102c957829003601f168201915b505050505081526020016003820180546102ff90610df0565b806  
01f016020809104026020016040519081016040528092919081815260200182805461032b90610df0565b801561037  
65780601f1061034d57610100808354040283529160200191610376565b820191905f5260205f20905b81548152906  
001019060200180831161023957829003601f168201915b50505050508152602001600482015f9054906101000a900  
473fffffffffffd57610100808354040283529160200191610376565b820191905f5260205f20905b81548152906001019  
fffffffd57610100808354040283529160200191610376565b820191905f5260205f20905b81548152906001019060201  
fffffffd57610100808354040283529160200191610376565b820191905f5260205f20905b81548152906001019060201  
fffffffd57610100808354040283529160200191610376565b820191905f5260205f20905b81548152906001019060201
```

```
If the transaction failed for not having enough gas, try increasing the gas limit gently.  
creation of IPProtection pending...  
view on Etherscan view on Blockscout  
[block:10246155 txIndex:29]  
from: 0xE10...f1323  
to: IPProtection.(constructor)  
value: 0 wei  
data: 0x608...10033  
logs: 0  
hash: 0xb44...a7707  
call to IPProtection.showIPTypes  
call  
[call]  
from: 0xe10ff59a60c18e1d8372f31d36fec442178f1323  
to: IPProtection.showIPTypes()  
data: 0x11d...3de74
```

```
0xE10Ff59a60C18E1D8372F31D36FE
c442178f1323
```

```
IPProtection.showIPTypes()  
0xB1A6fE9d16E321cAaBfA61f66B71  
eb9950fC007D
```

```
input          0x11d...3de74
               00000000000000000000000000000000
               09600000000000000000000000000000
               00001600000000000000000000000000
               00000000224000000000000000000000
               000000000000104832453280971161
output         01110116000000000000000000000000
               00000000000000000000000000000000
               013493245326711111212111410510
               31041160000000000000000000000000
               00000000000000000000000000000135
               032453284114971001011099711410
               70000000000000000000
decoded input  {}
               {
               "0": "string: 0 -
Patent",
decoded output "1": "string: 1 -
Copyright",
               "2": "string: 2 -
Trademark"
               }
logs          []
raw logs      []
```

```
>
```

```
[block:10246180 txIndex:28]
from: 0xE10...f1323
to: IPProtection.registerIP(string,string,string,uint8) 0xB1A...C007D
value: 0 wei
data: 0x7ec...00000
logs: 1
hash: 0x325...20a2b
status        1 Transaction mined and execution succeed
transaction hash 0x1932ad27c3ceccb9f0c622c6119c8d96ee0b7743e7cdb878c
               4c5aff1e632ac13
block hash     0x325d1e0c9e7dd17a3e186f035a9f4b0e0737dfedff3c7db17
               37f2cd93d120a2b
block number   10246180
from           0xE10Ff59a60C18E1D8372F31D36FEc442178f1323
to            IPProtection.registerIP(string,string,string,uint8)
               0xB1A6fE9d16E321cAaBfA61f66B71eb9950fC007D
transaction cost 187415 gas
decoded input  {
               "string_title": "mere bhole shankar",
```

decoded output

logs

```
raw logs
```

```
}  
]
```

[view on Etherscan](#) [view on Blockscout](#)

```
[block:10246183 txIndex:22]  
from: 0xE10...f1323  
to: IPProtection.registerIP(string,string,string,uint8) 0xB1A...C007D  
value: 0 wei  
data: 0x7ec...00000  
logs: 1  
hash: 0x068...31f79
```

```
status      1 Transaction mined and execution succeed  
  
transaction hash  0xbfe2abc052f0c1fe5412c90f4907b8536af080d0d3f9  
                  c7982923671d00ba49ee  
  
block hash      0x068c8af3c66b1bffe5b5be58ea3b569c904e785d7beea  
                  c0bf570febfb94731f79  
  
block number    10246183  
  
from            0xE10Ff59a60C18E1D8372F31D36FEc442178f1323  
  
to             IPProtection.registerIP(string,string,string,u  
int8)  
0xB1A6fE9d16E321cAaBfA61f66B71eb9950fC007D  
  
transaction cost 170315 gas  
  
decoded input   [{  
    "string_title": "mere bhole shankar",  
    "string_description": "shiv bhakti  
song",  
    "string_ipHash": "shiv12345",  
    "uint8_type": "1"  
}]  
  
decoded output  -  
  
logs           [{  
    {  
        "from":  
"0xB1A6fE9d16E321cAaBfA61f66B71eb9950fC007D",  
        "topic":  
"0x91f466b74e0c10f1b705ffflaf8faaa085958328064  
5fca9f02cf67c74a79fa5",  
        "event": "IPRegistered",  
        "args": {  
            "0": "2",  
            "1": "mere bhole shankar",  
            "2": "Copyright",  
            "3":  
"0xE10Ff59a60C18E1D8372F31D36FEc442178f1323"  
        }  
    }  
}]
```

[illegible]

```
>  
call to IPProtection.ipCount  
call  
[call]  
from: 0xe10ff59a60c18e1d8372f31d36fec442178f1323  
to: IPProtection.ipCount()  
data: 0x907...e67fe
```

[illegible]

[illegible]

Thank You

-NEELESHWARI MEHRA